



Fujitsu electronic paper with memory

ices, such as electronic paper, lightweight solar cells and inexpensive RFID tags. Compounded conductive polymers will play a critical role in the successful proliferation and miniaturisation of electronic devices, as they provide inexpensive protection against electrostatic discharge and electromagnetic or radio frequency interference.

Besides, conducting polymers have the potential to become alternatives to silicon-based electronics. It is believed that plastic electronics, based on thin-film transistors fabricated by organic films, will create a new range of products that could not be manufactured using conventional complementary metal oxide semiconductor (CMOS) approaches.

The flexibility of thin films also suggests a new product line, including roll-up displays, low-cost RFID tags, flexible sensor and photovoltaic arrays. Plastic electronics also generate less heat and use less power, thus alleviating two major problems bothering conventional electronics.

Graphene technology. New research findings suggest that use of graphene in microchips could make them faster in comparison to the standard silicon microchips. Thus, use of this technology in future cellphones and other communication systems will mean much faster transmission of data.

With conventional silicon-based

electronics, frequency cannot be increased, but graphene technology could help in developing systems in the 500 to 1,000 gigahertz range.

Graphene, a pure form of carbon, is the strongest material ever discovered. It also has a number of unique electrical properties, such as mobility—the ease

with which electrons can start moving in the material—which is 100 times that of silicon.

Researchers have already used the one-atom-thick layer of carbon atoms to make prototype transistors and other simple devices. Frequency multipliers are being widely used in radio communications and other applications. But existing systems require multiple components, signal filtering, and they consume more power. Whereas the graphene system has just a single transistor and produces a clean output efficiently.

Major challenges

In future, electronics has to tackle three major challenges:

Electronic waste. Electronics improve the way we live, work and play. Thus, it should have no impact on the environment. Through responsible use, reuse and recycling of electronic products, the electronics industry and consumers can protect and preserve the environment. The rapid pace of technological advancement in the field of electronics has made electronic gadgets both affordable and widely used. This results in large quantities of electrical and electronic equipment being added to the waste stream called electronic waste, also known as 'E waste.' It is generated when electronic products like computers, monitors, hard drives, copiers, printers, fax machines, mobile phones, telephones and televisions are

discarded. Electrical and electronic equipment are made up of a multitude of components, some containing toxic substances which can have adverse impact on human health and the environment. Often, these hazards arise due to the improper recycling and disposal processes used.

Green technology. Adopting green technology is about knowing what products to use, how to get best output from them, and what to do with them when they stop functioning. Best practices like how to stop wastage of energy, which gizmos are greener than others, and e-waste management are the prerequisites for moving towards green technology solutions.

Quality and quantity concerns.

Over the past many years, manufacturers have been devoting resources to fulfill the customers' requirements by changing designs and production parameters of electronic products, thus compromising on the quality of products. Increasing number of electronic products either discarded or being repaired before the minimum usage time indicate their poor quality.

Non-working central systems, problems related to data safety and streamlining of electronic systems are also serious problems faced by users. Electronic products are said to have more quality shortcomings than their mechanical counterparts. Billions of circuit boards enter the global market every year, and there are evidences to indicate that 5-10 per cent of the advanced circuit boards produced in small or medium batches require repairs in order to be approved for delivery under current IPC standards. Thus, quality and reliable products will play important role in shaping the future of electronics. We must also be able to produce them in quantity with reasonable market prices. There should be proper linkage between design, production and application. ●

The author is associated with the Department of Physics, Sant Longowal Institute of Engineering and Technology, Longowal, Sangrur, Punjab